The Bifröst

2025 CIV102 Project Team 202: Design Report

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1. Introduction

The purpose of this report is to document the design, analysis, and construction of our matboard bridge, *The Bifröst*. We named our bridge the Bifröst because we're fans of Thor: Ragnarok and we were inspired by the Bifröst's ability to span between worlds. Our bridge aims to connect our learning to real world design and show that even with limited materials good design can achieve remarkable strength and elegance.

Figure 1: Image of the Final Completed Bridge

The structure was intentionally developed with a simple and constructible geometry, while still maximizing structural efficiency under the CIV102 Matboard Bridge Design Project requirements. Our final design is a pi-beam bridge spanning 1200 mm between the centers of the two supports, capable of safely carrying the moving train load prescribed in the project brief.

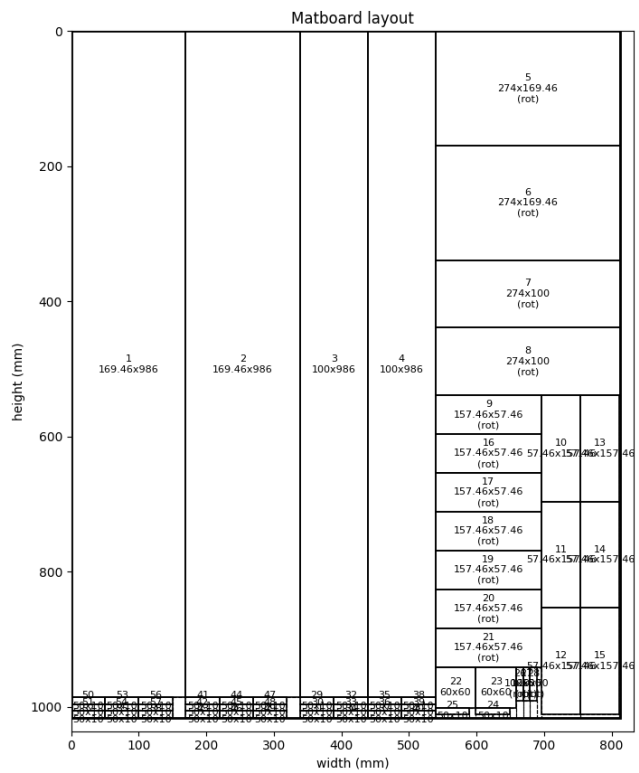


Figure 2: Image of the Final Matboard Cutout Schematic

The bridge was constructed from a single sheet of matboard of thickness 1.27 mm and two tubes of contact cement, fully adhering to the dimensional constraints on deck width, span length, and maximum allowable deck height. The resulting structure consists of a tall, slender box section with a deep web, a stiffened top compression flange, and a continuous flat deck that enables the passage of the train in both directions.

Our overarching goal was to correctly predict and maximize the predicted failure load while ensuring material efficiency and ease of fabrication. To achieve this, we began by analyzing the provided Design 0 cross-section to identify its structural weaknesses. Using insights from these calculations, we iteratively modified the cross-sectional geometry, diaphragm layout, and flange reinforcement strategy. At each iteration, we evaluated the structural performance using a numerical simulation script (implemented in MATLAB and Python), which generated the shear force and bending moment envelopes for a moving train (under a point load

assumption at both the train wheels and supports), computed updated sectional properties, and calculated factors of safety against all relevant failure modes, including flexural tension, flexural compression, matboard shear, glue shear, and both flexural and shear buckling of thin plates.

After applying this iterative process, we arrived at a final configuration: a pi-beam consisting of a 157.5 mm tall web, a reinforced double-layer 100 mm wide top flange, and 60 mm wide gap between the two webs. Diaphragms were placed at unequal positions along the flange. Starting from the left-most end of each support, the spacings were 12.5, 12.5, 91, 99, 110, 129, and 158.5 mm from the previous diaphragm location, respectively. This arrangement was guided by both the shear-force analysis and material-usage constraints. According to our simulations, the optimized design is predicted to fail by compression at an estimated train load of 1.367 KN.

Key Design Decisions and Iterations*:*

The progression from Design 0 to our final design was driven by quantitative analysis and practical construction considerations. Major design decisions included:

1. Doubling the top flange thickness to increase compression strength and delay local buckling.
2. Introducing more vertical diaphragms spaced along the span to maintain cross-sectional rigidity.
3. Increasing the overall bridge height to raise the second moment of area and reduce bending stresses.
4. Relocating longitudinal splices away from the midspan high-moment region to improve reliability.
5. Using unequal diaphragm spacing to reduce unnecessary material usage.
6. Varying the upper flange thickness along the span to place material only where bending demands are highest.

2. Design Iterations

a) <u>Design 0: Ground 0</u>

For iteration 0 (i.e. Design 0), our team decided to use the provided box-girder example as foundation of our entire project. We deemed that keeping the design simple would ultimately be our greatest strength, and decided against other structures, like trapezoidal or truss bridges, due to their lack of simplicity and difficulty in construction. The parameters and methods of which we developed the code to calculate for design 0 ultimately laid the foundations for our future iterations and enabled us to parameterize all the key components of our design. The key parameterization that we employed was determining the thickness of the top flange *t1*, the clear distance between the two webs *w2*, the total height of the bridge *h*, the glue tab length *w3*, and the spacing of the diaphragms *spacing*.

Through utilizing this code, we analyzed the forces from the train traversing the bridge at 1 mm increments along its span. From the code we determined that the critical failure mode for Load Case 1 was thin-plate buckling (Case 1) occurs approximately 44 mm from the center of the span, with a safety factor of only 0.597 (Figure 3). This location is the position of maximum bending moment. The analysis determined that the maximum load capacity of the beam is 257 N, which is substantially below Load Case 1. This inadequate factor of safety indicated that significant design optimizations were necessary to achieve acceptable performance.

```
Factors of safety for Design 0 under Load Case 1
FOS, flexual tension:       4.359
FOS, flexual compression:   1.037
FOS, glue shear:            7.479
FOS, matboard shear:        2.665
FOS, thin plate buckling 1: 0.597
FOS, thin plate buckling 2: 4.058
FOS, thin plate buckling 3: 4.720
FOS, shear buckling:        3.280
```

Figure 3: MATLAB code output for the different values of FOS at each failure mode for the bridge.

It is worth noting here, which will apply for the rest of the following iteration calcuations, several generalizations/overestimatings that our group did for the coding. To be careful not to overestimate the eventual failure of our bridge, for the contact of the glue to the top surface we assumed that to only be the length of the glue tab not the glue tab length plus the thickness of the matboard as well. This was done to reduce the effective area, considering the manufacturing error and uncertainties that are bound to occur, and to decrease our factor of safety estimate for the glue failure. Additonally we ignored the calculation of the shear failure of the glue for the connection of the two segments of the top flanges (double layer) because from the equation:

$$\tau = \frac{V}{A}$$

we know that the failure due to glue will be much higher than the failure due to glue at the glue tab, based on the increased thickness therefore it is negligible. As well, when calculating case 1 of thin plate buckling stress, we took *b* as the edge-to-edge distance between the walls for a more conservative estimate.

$$\sigma = \frac{4\pi^2 E}{12\left(1-\mu^2\right)}\left(\frac{t}{b}\right)^2$$

The glue tabs were ignored here since compressive forces act mostly on the web panels themselves. For Case 3 shear buckling, we only considered the tall web segment because this panel is so long and slender that it will control over the other, much shorter sections. Finally, for the thin-plate shear buckling, we subtract the width of the glue tab in our estimation of the shear buckling for the t values.

$$\tau\ buckling = \frac{5\pi^2 E}{12\left(1-\mu^2\right)}\left(\left(\frac{t}{spacing}\right)^2 + \left(\frac{t}{(h-t1-t)}\right)^2\right)$$

In the code, the Design 0 geometry was encoded through the definition of the top flange area A1, web areas A2, bottom flange A3, and glue-tab flange A4, along with their corresponding centroidal distances. These feed into the centroid computation *ybar* and second moment of area (I) using the parallel-axis theorem. The code segment containing the definitions of A1–A4, d1–d4, and I1–I4 effectively decomposes the cross-section into four rectangles whose combined structural properties form the baseline of our analysis. These section property results are illustrated conceptually in Figure 4, which corresponds directly to the geometric quantities used in our computational model.

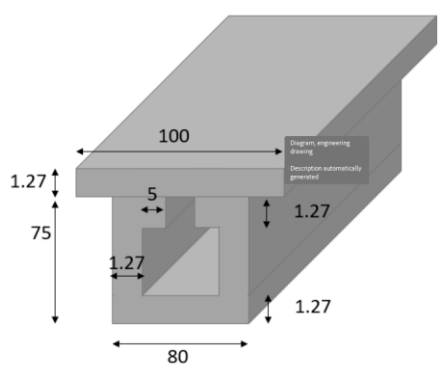


Figure 4: Cross-sectional schematic of the Design 0 box-girder, highlighting top flange, webs, bottom flange, and glue-tab

With the baseline geometry defined, we then subjected the bridge to the full moving-load envelope analysis required for Load Case 1 (400 N train). In the code, the span was discretized into 1 mm intervals, and the six axle loads were shifted across the bridge from the front axle at x = 0 mm until the rear axle reached x = 1200 mm, producing a total of L − train_length + 1 load

cases. For each of these train positions, the script computed the support reactions using static equilibrium, assembled a load vector containing all axle loads and support forces, and calculated the corresponding shear force and bending moment using cumulative numerical integration. By extracting the maximum absolute values across all load cases, the script produced the shear force envelope *SFD_env* and bending moment envelope *BMD_env*. These envelopes as shown in Figure 5 represent the maximum shear and moment carried when the train moves along the bridge.

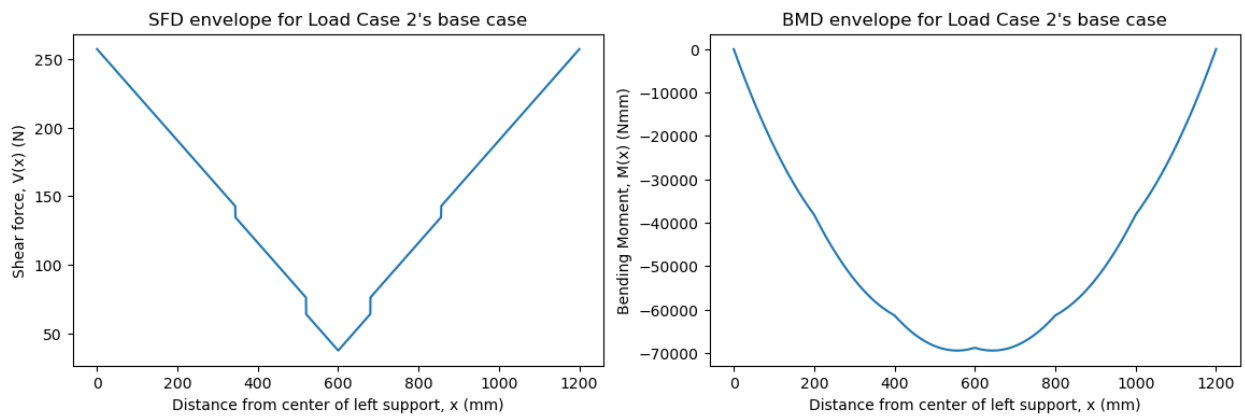


Figure 5: Plot of Design 0 shear and moment envelopes across the span.

Using the maximum moment, Mmax, and maximum shear, Vmax, from these diagrams, we computed the stresses in the Design 0 cross-section. The flexural stresses at the extreme fibers were obtained through:

$$S_{\text{top}} = \frac{M_{\text{max}} y_{\text{top}}}{I}, S_{\text{bot}} = \frac{M_{\text{max}} y_{\text{bot}}}{I},$$

And matboard shear and glue shear stresses were calculated using:

$$T_{\text{cent}} = \frac{V_{\text{max}} Q_{\text{cent}}}{I\, t_{\text{cent}}}, T_{\text{glue}} = \frac{V_{\text{max}} Q_{\text{glue}}}{I\, t_{\text{glue}}}.$$

Tension, compression, shear and thin-plate buckling limits for each flange and web panel were then evaluated, allowing the script to compute factors of safety against flexural tension, flexural compression, matboard shear, glue shear, flange buckling (buckling cases 1–3), and shear buckling of the webs.

The analysis revealed that the governing failure mode for Design 0 is buckling case 1, buckling of the unsupported central portion of the top flange which occurd at the point of maximum bending moment near midspan (approximately x ≈ 644 mm). The predicted factor of

safety for this mode was 0.597 and the compressive stress in the top flange was also close to its limit with FOS comp equal to 1.037 Other modes of failure had significantly higher factors of safety and were not critical for. The smallest FOS across all failure modes led to a predicted failure load of only 255 N, far below the 400 N train load. This confirmed that the baseline section was substantially under-designed and that multiple structural improvements were required.

Failure Mode	***Before (Design 0)***
FOS_tens	4.36
FOS_comp	1.037
FOS_shear	2.66
FOS_glue	7.48
FOS_buck1	0.597
FOS_buck2	4.06
FOS_buck3	4.72
FOS_buckV	3.28
Controlling Mode	**FOS_buck1**
Predicted Failure Load (N)	**260**

Table 1: Failure Mode Comparison for Design 0 and Design 1

Governing Assumptions Troughout the Design Process:

During this initial analysis, we also introduced several conservative modeling assumptions that carried forward into our future iterations. For glue shear, we assumed load transfer only through the length of the glue tab at the top flange and not through any additional matboard thickness in contact, which reduces the effective glue shear perimeter and therefore lowers the predicted FOS. For double-flange configurations introduced later, we did not compute shear transfer between the flange layers, since that interface is significantly stronger than the governed glue-tab interface and would not realistically control failure. In thin-plate buckling calculations, we consistently assumed the most extreme distances (like the outside web panel widths) as our b value in the equation $\sigma_{cr} \propto (t/b)^2$. We also focused the buckling case 3 calculation on the longest, slender wall segment only, because its high slenderness ratio makes it the dominant controlling panel. Finally, for the shear buckling formula:

$$T_{\text{buck}} = \frac{5\pi^2 E}{12(1-\mu^2)}\left[\left(\frac{t}{\text{spacing}}\right)^2 + \left(\frac{t}{h - t_1 - t}\right)^2\right],$$

we used the reduced clear web height $h - t_1 - t$ (intentionally ignoring the t2 thickness of the web) to avoid unintentionally increasing the shear buckling capacity. These conservative assumptions ensured that any improvement in factors of safety later in the design process corresponded to real structural enhancement rather than optimistic modeling.

b) <u>Design 1: Adding Top-Flanges</u>

It is worth noting that after Design Iteration 0, we implemented a Python script that would automatically sort the designed members, flanges, and walls to optimize their arrangement onto the matboard for a cutout layout. This code optimizes how to cut a set of rectangular parts from a large sheet by simulating the placement of each part (with optional rotation for better fit) and updating the open space after each part is placed. It tests different ways to splice long pieces so that all required parts fit within the given sheet dimensions, printing and plotting a feasible solution if one is found.

To pre-screen any implementations or designs that would fail the above code, we implemented a volume checker that automatically returns a max load of 0 if it calculated that the volume of the cross section across our entire bridge exceeded the volume of the provided matboard.

We realized that Design 0 would fail immediately to Shear Buckling Case 1, with the FOS being under 1, with a peak failure of 259.5 N (under load case 1). We realized that the second moment of area for the initial design was far too low, resulting in excessive stress in the top flange (stress = M * y / I), caused by the very low second moment of area value. To remedy this, **we chose to include a second layer for the top flange** to increase the second moment of area. This in turn would decrease the maximum compressive stress of the top flange and increase all of the FOS values, particularly for the case 1 buckling, which jumped from 0.597 to 4.10, improving the maximum loading from 259.45 to 667.746 N. While the overall maximum loading value has increased through the addition of this top flange, the newest limited feature has become compressive failure.

It is worth noting that our code considers the subsequent passes for load case 2, where each consecutive freight train has a value 1.1 times larger than the previous pass, and the locomotive at the very front has a mass 1.35 times larger than the middle train.

The following table compares the factors of safety (FOS) for all relevant failure modes. The naming conventions correspond to tension (FOS_tens), compression (FOS_comp), shear (FOS_shear), glue shear (FOS_glue), the three flange buckling cases (FOS_buck1–3), and web shear buckling (FOS_buckV).

Failure Mode	***Before (Design 0)***	***After (Design 1)***
FOS_tens	4.36	4.70
FOS_comp	1.037	1.669
FOS_shear	2.66	2.58
FOS_glue	7.48	5.91
FOS_buck1	0.597	4.10
FOS_buck2	4.06	26.1
FOS_buck3	4.72	14.83
FOS_buckV	3.28	3.50
Controlling Mode	**FOS_buck1**	**FOS_comp**
Predicted Failure Load (N)	**260**	**668**

Table 1: Failure Mode Comparison for Design 0 and Design 1

c) Design 2: Optimizing Top-Flange Layering and Height

After improving the predicted peak loading to 668 N, we realized that the bridge design was now limited by the compressive force and the maximum shear that the bridge cross section could sustain. Our solution to this issue, after adding an additional layer to the top flange, was to increase the overall height of the bridge. By increasing the height, we would effectively increase the second moment of area (I) value; with more mass located farther from the centroidal axis, the parallel-axis theorem dictates that I increase in a square relation. However, at this point in the process, we realized that the height and the number of layers for the top flange were becoming conflicting variables. If we increased the height, we would have less material for additional layers and vice versa.

Additionally, through our python algorithm, which checks whether proposed designs are physically possible given the matboard dimensions, it became increasingly clear that an optimization of the height and the number of layers for the bridge would be necessary. It is worth

noting that we could generally ignore the effects of increasing the centroidal axis location for the bridge, as the FOS for the tension (i.e. the distance from the centroidal axis to the bottom of the cross section) was approx. 4.7 (as a result from the very large tensile shear capacity (30 MPa) that the matboard material has). As a result, in our optimization we focused on iterating through designs that improved the compressive and shear characteristics of the bridge itself.

At this point, we used our MATLAB script to crudely check whether a cross section was feasible. The code iterated through all reasonably potential values for the number of top-flange layers (1-7) and the height of the cross-section (20-180 mm in 20 mm increments). For a design that violated material limits, the maximum load was automatically set to 0N. Using these criteria, we calculated that the maximum potential configuration that our material constraints yielded was **3 top-flange layers at 180 mm height** (maximizing the minimum FOS for compression and shear). From the Design 0, we were able to record a 346% increase in the minimum FOS of the bridge, an increase from 1.67 to 5.78.

Further evidence of successful optimization can be observed from the differences between the compressive and shear FOS values. The compressive and shear values 6.01 and 5.78 respectively indicate that the optimized design was incredibly effective to within 3.98% of one another. While the overall design peak load decreased to 622 N, we were able to optimize the height and the number of top flange layers, to basically finalize our second moment of area design. It is also worth noting that, from this iteration onward, a new limiting mode emerged: thin-plate buckling. This occurred because, since Iteration 0, no changes had been made to the diaphragm placement along the bridge (i.e., the diaphragms were still spaced 400 mm from one another). This lack of bracing increased the slenderness of the webs, making thin-plate buckling the new governing failure mechanism.

Failure Mode	***Before (Design 1)***	***After (Design 2)***
FOS_tens	4.70	16.33
FOS_comp	1.669	6.01
FOS_shear	2.58	5.78
FOS_glue	5.91	13.94
FOS_buck1	4.10	33.2
FOS_buck2	26.1	212
FOS_buck3	14.83	9.34

FOS_buckV	3.50	1.556
Controlling Mode	**FOS_comp**	**FOS_buckV**
Predicted Failure Load (N)	**668**	**622**

Table 2: Failure Mode Comparison for Design 1 and Design 2

d) Design 3: Decreasing Diaphragm Spacing

As described in Design 2, we were able to improve and optimize the heights of the bridge as well as the layering of the top flange to protect against compressive and shear failures. Now, the restricting mode to our bridge is the thin-plate shear buckling at 1.556 as demonstrated in Table 2. At this point in the iterative process, no changes have been made to the initial design regarding the plate spacing. Thus, we began brainstorming ways to improve this value. Since the thin-plate buckling equations are affected by both the diaphragm heights and the separations between each plate, we determined that it would be unwise to alter the geometry of the existing cross-section after already optimizing the height and plate thicknesses. Instead of modifying the section again in hopes of further improving the second moment of area or reducing diaphragm size, we focused on adjusting the diaphragm spacing itself. Thus, we decided to **decrease the spacing from the initial 400 mm to 100 mm** (randomly chosen just to observe the impact of this decision on the design itself, in future designs this value will be optimized). After making this change, we were able to improve our maximum load capacity by 3.5 times to 2.12 KN. As demonstrated in Table 3; after making this change we were able to increase the plate buckling FOS to 5.3. After this change we can observe that every single FOS for each mode we are considering is well above 5, indicating an extremely strong bridge design.

However, to attain this incredibly large FOS value, we completely ignored the reality of the design and the manufacturability of our bridge. For the purposes of our initial iterations, we even neglected the volume-conservation requirement outlined in Design 0. In subsequent designs, we began to recognize the difficulty and the challenges involved in developing the strongest possible bridge that can still be physically constructed with the materials and geometry available to us. Therefore, the next iterations focus on minimizing material usage to achieve a feasible and manufacturable bridge.

It is worth noting that while we assume the diaphragm spacings are generally reduced to a single value (e.g., 100 mm), what we also mean is that this symmetric placement of diaphragms begins at the center of the bridge. This ensures a strong foundation where the highest density of bending moment occurs (the maximum moment is about 35 mm to the right of the exact center of the bridge under load case 2), and where deformations from shear-plate buckling will initiate.

As well, two diaphragms are located at the center of the supports on both sides to protect against thin-plate buckling, as well as to ensure increased distribution of the support reaction forces to the bridge webs and top flange.

Failure Mode	***Before (Design 2)***	***After (Design 3)***
FOS_tens	16.33	16.33
FOS_comp	6.01	6.01
FOS_shear	5.78	5.78
FOS_glue	13.94	13.94
FOS_buck1	33.2	33.2
FOS_buck2	212	212
FOS_buck3	9.34	9.34
FOS_buckV	1.556	5.30
Controlling Mode	**FOS_buckV**	**FOS_buckV**
Predicted Failure Load (N)	**622**	**2120**

Table 3: Failure Mode Comparison for Design 2 and Design 3

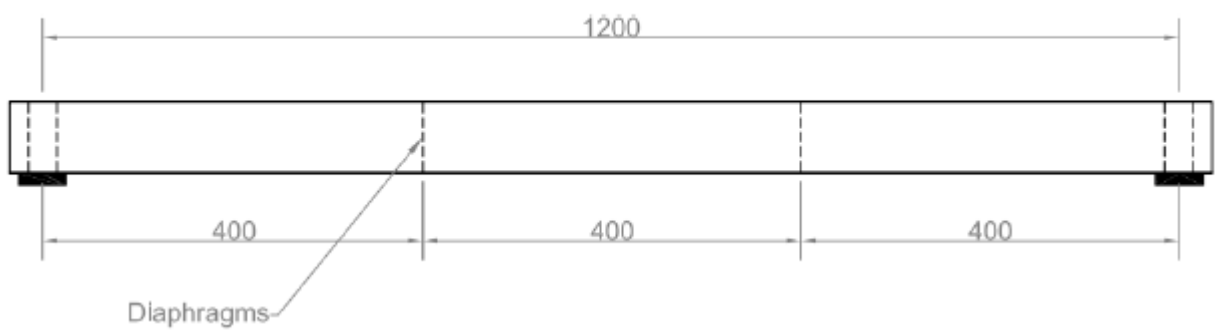


Figure 6: Image Of Diaphragm Placement for Design 0

e) Design 4

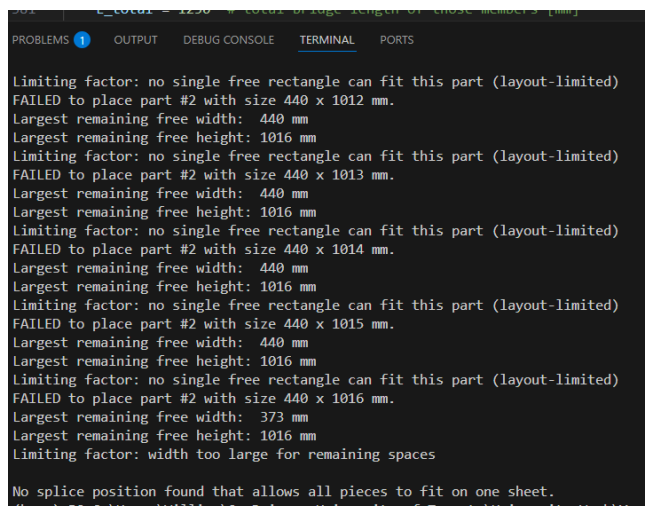
As described in the Iteration 3 we were able to design and simulate *The Bifrost* for a peak load of 2.12 KN. However, this design was unable to be validated as a feasible design that is manufacturable. As a result, in this iteration we focused on increasing the realistic nature and generating a methodology and procedure to check if a design configuration is feasible or not.

The first step in this process was developing a program to simulate for a series of rectangular shapes, whether they can all fit within a larger rectangle. This program worked through exhausting all potential orientations and placement locations for every single rectangle utilizing a depth first search like algorithm.

The second step in this process was realizing that our bridge length of >1200 mm would exceed the longest length of our matboard paper (1016 mm). We determined that our bridge would require splicing of key elements (for the top-flange, bottom-flange, and webbing) to ensure that we are able to fit all our elements onto a single matboard schematic design.

Through a combination of our first and second steps, we were able to determine the methodology for determining which design geometries were functionally impossible. If we checked all splicing locations and our fitting board function was unable to find a single potential matboard fitting geometry, we would know that a design is functionally useless. On the other hand, if we were able to find a splicing location, then we would know that our design was manufacturable and feasible.

After putting our Design 4 through the above methodology, we determined with confidence, that our previous iteration would be ultimately impossible to manufacture as demonstrated by the code output in Fig.6. To remedy this issue, we began devising methods to reduce our effective area and hopefully find a geometry which enables all components to fit within. This led to a series of rapid iterations all under the guise of reducing the effective area while maintaining the highest potential loading force, to get a bridge that is possible to manufacture.



```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Limiting factor: no single free rectangle can fit this part (layout-limited)
FAILED to place part #2 with size 440 x 1012 mm.
Largest remaining free width:  440 mm
Largest remaining free height: 1016 mm
Limiting factor: no single free rectangle can fit this part (layout-limited)
FAILED to place part #2 with size 440 x 1013 mm.
Largest remaining free width:  440 mm
Largest remaining free height: 1016 mm
Limiting factor: no single free rectangle can fit this part (layout-limited)
FAILED to place part #2 with size 440 x 1014 mm.
Largest remaining free width:  440 mm
Largest remaining free height: 1016 mm
Limiting factor: no single free rectangle can fit this part (layout-limited)
FAILED to place part #2 with size 440 x 1015 mm.
Largest remaining free width:  440 mm
Largest remaining free height: 1016 mm
Limiting factor: no single free rectangle can fit this part (layout-limited)
FAILED to place part #2 with size 440 x 1016 mm.
Largest remaining free width:  373 mm
Largest remaining free height: 1016 mm
Limiting factor: width too large for remaining spaces

No splice position found that allows all pieces to fit on one sheet.
```

Figure 7: Screenshot of Fitting Design 3 Onto Matboard

4.1: Removing the bottom flange

After careful consideration and deliberation, we realized that our bottom flange for our bridge was ultimately unnecessary and decided to remove it entirely. Doing this would enable us to utilize much more material and space on the matboard. We justified this because as demonstrated in Table.4 before rem**oving the bottom flange**, we had an extremely large FOS bucking case 2 (i.e. the overhanging section of the bridge). Effectively this was wasted material, and an overestimation on our part for that section of the bridge. The same could be said for the cross-section FOS for tension, as well as the buckling case 3. We realized that by removing the bottom flange we would be able to reduce the FOS for tension, compression, shear, glue, case 1 buckling, and thin-plate buckling. While removing the bottom flange effectively reduced our maximum load by about 12%, this was well justified as we were able to save approximately 25 percent of the total material. As a result of the removal of the bottom flange, our design has turned into a pi-beam. As shown in Fig.5, from the fitting code output, we were still unable to fit all the design within the matboard.

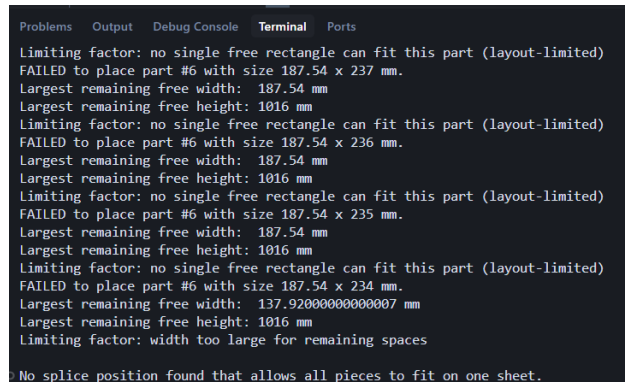


Figure 8: Screenshot of Fitting Design 4.1 Onto Matboard

Failure Mode	***Before (Design 3)***	***After (Design 4.1)***
FOS_tens	16.33	9.63
FOS_comp	6.01	5.03
FOS_shear	5.78	5.07
FOS_glue	13.94	11.78
FOS_buck1	33.2	27.8
FOS_buck2	212	177.4
FOS_buck3	9.34	13.1
FOS_buckV	5.30	4.65
Controlling Mode	**FOS_buckV**	**FOS_buckV**
Predicted Failure Load (N)	**2120**	**1861**

Table 4: Failure Mode Comparison for Design 3 and Design 4.1

4.2: Decreasing the distance between the two walls. Making the clear distance smaller

The next step to decreasing the effective area was to minimize the clear distance between the two walls of the cross section. Up until this point, this value was still set at the initial 80 mm. However, after reviewing all the failure modes, it became clear that this value was not fully optimized. This effect can be observed when analyzing the difference in FOS values between buckling case 1 and case 2. We can clearly see a massive difference between the two, which indicates that the distance at which buckling case 1 and buckling case 2 occur is not optimized for equal FOS values. Thus, this demonstrates a facet that remains unoptimized within our bridge design. To remedy this, we implemented another MATLAB program to optimize the value of w2 (the distance between the two webs), simulating every possible value of w2 from the total top-flange height down to 0 and finding the smallest difference between the FOS_buck1 and FOS_buck2 values.

As it turns out, the optimal **w2 value is approximately 60 mm**, at which the FOS_buck1 and FOS_buck2 difference is approximately 2.20, a massive improvement from the original difference of 149.6. While Design 4.2 does not change the max loading, nor the material saved (as the bottom flange has now been removed) it serves as an indicator for the positioning of the two walls from one another and we believe this iteration is necessary to prevent failing in unpredictable manners, in this way we can ensure that the FOS values are very close (as shown in Table 5). It is worth noting that this design was unable to pass the Python fitting algorithm.

Failure Mode	***Before (Design 4.1)***	***After (Design 4.2)***
FOS_tens	9.63	9.63
FOS_comp	5.03	5.03
FOS_shear	5.08	5.08
FOS_glue	11.78	11.78
FOS_buck1	27.8	48.85
FOS_buck2	177.4	46.7
FOS_buck3	13.14	4.65
FOS_buckV	4.65	4.65
Controlling Mode	**FOS_buckV**	**FOS_buckV**

Predicted Failure Load (N)	**1861**	**1861**

Table 5: Failure Mode Comparison for Design 4.1 and Design 4.2

4.3: Remove the top flange thickness by 1. Decrease the height by 20.

We realized that for our current design, since it was still unable to pass the matboard fitting algorithm, it would be unrealistic to maintain the same high expectation of a 1 kN peak loading, and we decided to lower our expectations. As a result, our next decision was to sacrifice some strength and decrease the thickness of the top flange from the previous 3 layers to only 2 layers and reduce the height from 180 mm to 160 mm, to enable the full design to fit within the given matboard dimensions. Consequently, our second moment of area values decreased, and the final limiting mode returned to compression, with an FOS of 3.30, resulting in an effective maximum load of 1.319 kN. However, by sacrificing this strength, we gained the ability to actually build the bridge. As shown in Fig. 7, the fitting algorithm succeeded, and all components of our bridge were finally able to fit onto the matboard.

Through this iterative process of reducing area and optimizing dimensions within the fourth design iteration, we were able to conceptualize and imagine our bridge as a simple pi-beam with a height of 160 mm, a top-flange width of 100 mm, a 60 mm spacing between the webs, 2 layers on the top flange, and diaphragm spacing of 100 mm. For this configuration, our bridge will still fail due to compression, with a peak load of 1.319 kN. However, it is worth noting that the shear failure is also very close, meaning this is a dangerous value and something worth monitoring in future iterations.

Failure Mode	***After (Design 4.2)***	***After (Design 4.3)***
FOS_tens	9.63	7.34
FOS_comp	5.03	3.30
FOS_shear	5.08	4.55
FOS_glue	11.78	11.42
FOS_buck1	48.85	14.22
FOS_buck2	46.7	13.60
FOS_buck3	4.65	8.33
FOS_buckV	4.65	4.43
Controlling Mode	**FOS_buckV**	**FOS_buckV**

Predicted Failure Load (N)	**1861**	**1319**

Table 6: Failure Mode Comparison for Design 4.2 and Design 4.3

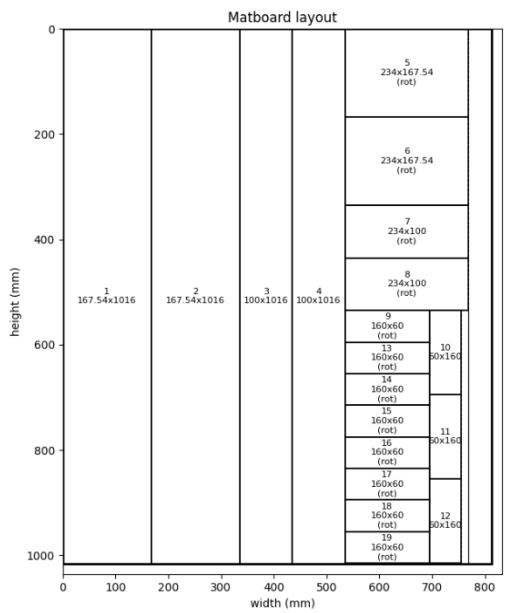


Figure 9: Screenshot of Fitting Design 4.3 Onto Matboard

f) <u>Design 5</u>

The next optimization we attempted to reduce the effective area used was to vary the diaphragm spacing and place them at unequal distances from one another. We were able to optimize the placement of the diaphragms using MATLAB code. This was achieved by applying several boundary conditions and a few clever generalizations. For example, the algorithm enforced that each support must have two diaphragms, the center of the span must contain a diaphragm, and the overall spacing pattern must remain symmetric about the center. The code then accepted a target FOS and generated the minimum number of diaphragms, along with their corresponding spacings, required to achieve that target.

Specifically, the algorithm used the shear envelope (the maximum shear at every x-location) together with the shear thin-plate buckling equation to compute the minimum allowable spacing at each position along the bridge. Using this array of minimum permissible spacings, we then generated the optimized diaphragm locations. The procedure was as follows: starting from

the pair of diaphragms at each support, the code added the spacing value from the spacing array to determine the location of the next diaphragm, repeating this process until the diaphragm reached or exceeded the center of the span. The resulting pattern was then mirrored on the opposite side.

From this method, we calculated that for a target thin plate buckling FOS of 5.8 (comfortably exceeding all the other FOS values from Design 4), the minimum number of diaphragms required is 9 (excluding the center diaphragm and not counting the double diaphragms at the supports). This results in a total of 13 diaphragms placed at spacings of 12.5, 12.5, 91, 98, 110, 129, and 158.5 mm. With only 13 total diaphragms, we were able to achieve an FOS of 5.8 before a 14th diaphragm would have been required, exceeding our available matboard space.

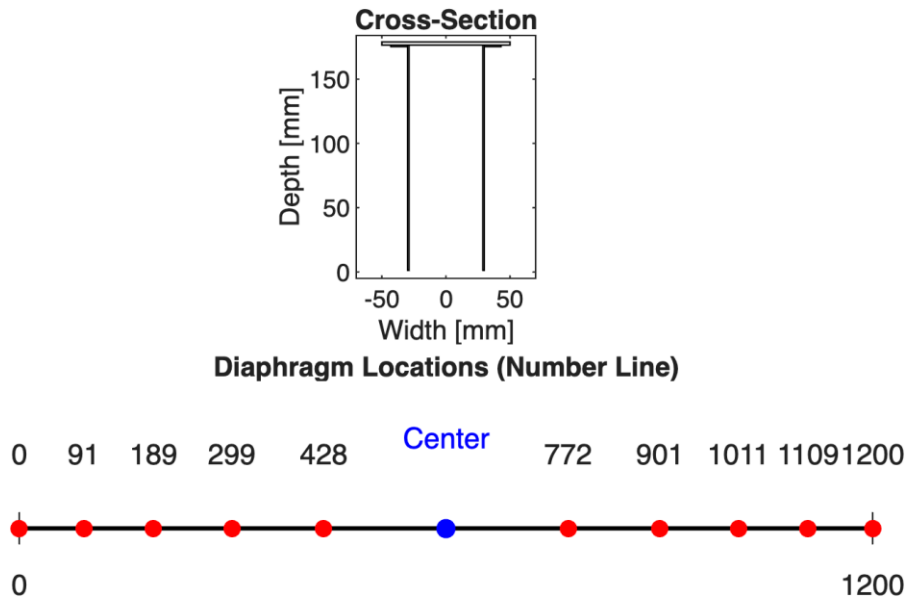


Figure 10: Code Output For The Unequal Diaphragm Placement

g) <u>Design 6</u>

The final design iteration focused on optimizing the thickness of the top flange at different locations down the bridge. We realized that it was unnecessary to put all layers of the top-flange reinforcement evenly along the entire bridge. As the compressive and shear effects vary throughout the entire bridge, it becomes unnecessary to design for the peak compressive and shear FOS for all locations. Thus, to overcome this, a MATLAB program was constructed to optimize a two-zone top flange layout under a total strip-length budget. As this description implies, we input a specific length of available top flange material into the program, and according to a two-zone layout (i.e. longer lengths in the middle for the larger moments, and the

thinner flanges at the ends of the bridge). The program outputs the optimal lengths of each segment as well as the number of layers that would result in the largest minimum FOS for each configuration. To do so, the program checks compression and shear FOS in both the edge sections, and stores the smallest values, and after checking through all the possible options, the best configuration yielding the largest FOS will be output to the user.

To interpret these results, the L_edge value is the length of the outside segment, and the L_mid is the length of the middle segment. Interestingly if you only have a whole integer of the total length of the bridge available, i.e. n = 1200 the optimal configuration will be a bridge with a single top-flange thickness (i.e. 600 for the L_edge and 0 for the L_mid). However, when you input a value which is a little bit longer than the length of the bridge itself, it will output a variant and the optimal configuration. As demonstrated in Fig.9 below which indicates the varying outputs for the varying lengths. Therefore, for our available materials, the best configuration is the entire top flange being composed for the exact same length of 2 layers.

```
max_top_length = 1400

top_opt = struct with fields:
            n_edge: 1
             n_mid: 2
           t1_edge: 1.2700
            t1_mid: 2.5400
             L_mid: 200
            L_edge: 500
            L_used: 1400
     FOS_comp_edge: 2.6009
      FOS_comp_mid: 3.8294
    FOS_shear_edge: 5.0884
     FOS_shear_mid: 5.1010
           FOS_min: 2.6009
```

```
max_top_length = 2400

top_opt = struct with fields:
            n_edge: 2
             n_mid: 2
           t1_edge: 2.5400
            t1_mid: 2.5400
             L_mid: 0
            L_edge: 600
            L_used: 2400
     FOS_comp_edge: 3.8294
      FOS_comp_mid: 3.8294
    FOS_shear_edge: 5.1010
     FOS_shear_mid: 5.1010
           FOS_min: 3.8294
```

Figure 11: Code Output For The Unequal Thickness For 1400 and 2400 mm Top Lengths

3. Conclusion

The design and construction of *The Bifröst* demonstrated the importance of structural analysis, material constraints, and construction to create a successful bridge. From the start, our goal was to accurately predict the failure load while keeping the design simple and easily buildable. As we progressed through each iteration, it became clear that small changes to the geometry affected multiple aspects of the bridge's behaviour. Increasing the overall height increased the second moment of area but made the webs more susceptible to shear, reinforcing the top flange improved the resistance to compression but required more material and adjusting glue-tab dimensions influenced the shear. Because of this, we treated the optimization of the design as a system and recognized the impact of attempting to optimize any single parameter on its own.

The final design is a 160 mm tall pi-beam with a 100 mm double-layer top flange, two centered webs spaced 60 mm apart, and an unequal diaphragm layout. This overall design maximized the performance of the bridge while balancing materials, to ensure that the design was able to fit onto the entire matboard surface.

Although our analytical approach relied on theoretical formulas which ignored practical factors such as glue variability, cutting accuracy, The combination of hand calculations, MATLAB and Python code allowed us to determine that the predicted failure mode would be compression and *The Bifröst* would have capacity of 1.367 kN. Overall, the project showed us the importance of understanding how each design decision affects bending, shear, and buckling. The creation of *The Bifröst* gave us valuable experience in applying engineering analysis to a real structure.

4. Bibliography

[1] A. Kuan and M. P. Collins, "CIV102 – Structures and Materials: An Introduction to Engineering Design," Course Notes, University of Toronto, Division of Engineering Science, Toronto, ON, Canada, Aug. 2022.